

Sri Datta Ganesh Bandreddi

AI Engineer | agentic LLM systems, retrieval, and evidence-grounded applications

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EDUCATION

CARNEGIE MELLON UNIVERSITY

Master of Science in AI Engineering, Biomedical Engineering | GPA: 4.00/4.00

Pittsburgh, PA

Dec 2026

VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY

Bachelor of Technology, Mechanical Engineering | GPA: 8.44/10.00

Nagpur, India

Jul 2022

SKILLS

Languages: Python, C++, Go, Java, SQL, Bash /Shell Scripting

GPU (Acceleration): CUDA, cuDNN, cuBLAS, TensorRT, Triton Inference Server, NCCL, Nsight Systems/Compute

DL Frameworks: PyTorch, JAX, TensorFlow, HF, NeMo/Megatron-LM, LoRA/QLoRA, FP8/BF16 Mixed Precision

Data & ETL: Rapids (cuDF, cuML), NVTabular, DALI, NumPy, SciPy, Pandas, SparkML

Systems & MLOps: Docker, Kubernetes, Slurm, Jenkins/GitHub Actions, MLflow, Git, AWS (ECS, SageMaker), GCP

Data Infrastructure: PostgreSQL, Neo4j, Kafka, Redis, Iceberg, Elasticsearch, FAISS, MongoDB

PROFESSIONAL EXPERIENCE

MOTIONAL

Pittsburgh, PA

Data Platform Intern (Infrastructure)

May 2026 - Aug 2026

- Delivered evidence-driven natural-language drivelog search, by building an agentic LangGraph retrieval service on FastAPI that collates metadata and alternate source-of-truth signals into grounded, tool-calling answers
- Cut drivelog thaw costs by about \$100K/month, by pairing the retrieval agent with a live-storage metadata layer so common questions are answered without thawing logs from cold storage
- Built a retraining flywheel for AV models and earned a provisional patent, by having the agent select and timebox-thaw long-tail events over a scene-embedding retrieval index

PHENOM

Hyderabad, India

Product Development Engineer (Enterprise Talent Graph)

Dec 2023 - Jul 2025

- Led a team of six engineers to re-architect a multi-tenant AI analytics platform, enhancing scalability and reliability under concurrent tenant load by refactoring shared services and reducing cross-tenant coupling
- Scaled production knowledge graph to 1M+ entities with continuous ingestion, by designing validated aggregation pipelines integrated with Neo4j graph schemas and Iceberg versioned storage
- Ensured high availability for production workloads serving 100+ enterprise clients, by managing distributed Kafka, PostgreSQL, and Redis clusters and building CI/CD pipelines using Jenkins and Helm charts on Kubernetes
- Accelerated AI-driven query response times by 40% across 100+ clients, by tuning PostgreSQL horizontal sharding strategies

BYJU'S

Hyderabad, India

Software Engineer (Core Platform Team)

Jul 2022 - Dec 2023

- Cut infrastructure costs by 70% while maintaining reliability for 1M+ users, by deploying scheduled AWS ECS auto-scaling policies and maintaining Infrastructure-as-Code (IaC) configurations based on user traffic patterns
- Eliminated dependency on external identity services by 50%, by leading development of an in-house authentication and authorization platform using Go and Java (Spring Boot)

PHENOM

Hyderabad, India

Machine Learning Intern

May 2021 - Jun 2022

- Boosted recommendation accuracy by building persona classification models using NLP techniques (spaCy, BERT embeddings) across 3 data sources, processing user behavioral signals at scale to serve personalized content for production recommendation pipelines
- Accelerated recommender system feature preprocessing by migrating CPU-based data pipelines to NVTabular on GPU, reducing preprocessing latency and validating scalability improvements across 2 production databases

PROJECT EXPERIENCE

Research Project, Efficient Attention Mechanisms (Carnegie Mellon University) (link)

Aug 2025 - Present

- Validated over 200x speedups and sub-quadratic memory scaling for linear attention mechanisms, by implementing and benchmarking on Long Range Arena tasks at 1k-8k token sequence lengths
- Profiled multi-GPU training with NCCL across FP8/BF16 configurations using Nsight Systems on Slurm-managed clusters, identifying optimal precision-performance tradeoffs for attention variants
- Co-authored the linear-attention operator (3rd of 7 authors, submitted to TMLR), later adopted by an NVIDIA and General Motors crash-dynamics effort for about 2x lower memory, across 5+ efficient attention architectures benchmarked on vision and diffusion tasks

CT-Transformer, Vision Transformers (Carnegie Mellon University)

Sep 2025 - Dec 2025

- Achieved 96.22% macro AUC-ROC on 15-class VinBigData chest X-ray classification outperforming ResNet-50, EfficientNet-B3, and ViT baselines, by training a Swin Transformer scaled to 210M parameters with fused attention kernels
- Optimized inference latency by 3x over PyTorch eager execution, by converting trained models to TensorRT (FP16/INT8), accelerating preprocessing with DALI, and deploying on Triton Inference Server with adaptive request batching

Distributed LLM Fine-Tuning with Megatron-LM (Carnegie Mellon University)

Aug 2025 - Oct 2025

- Fine-tuned a 7B parameter LLM with stable convergence and about 40% reduced memory footprint on 4x H100 GPUs, by configuring tensor and pipeline parallelism with NCCL-backed communication, gradient checkpointing, and using LoRA/QLoRA adapters
- Achieved 2.1x training throughput improvement over naive data parallelism, by tuning NCCL parameters, gradient bucketing, and optimizer state sharding across GPUs

AWARDS & HONORS

Phenomenal Award for Outstanding Performance, Phenom (2024, Q1 & Q2) • 2nd Place, AppLovin Track, TartanHacks 2026 • Provisional Patent, Motionai (2026)